# Moving does not require an unbalanced force

**GUIDED NOTES / use with the projected diagram** Name _____

## A SAY THE WORD

### Inertia / ES: inercia / PT: inércia

An object keeps its velocity unless a net force changes it. More mass, more inertia.

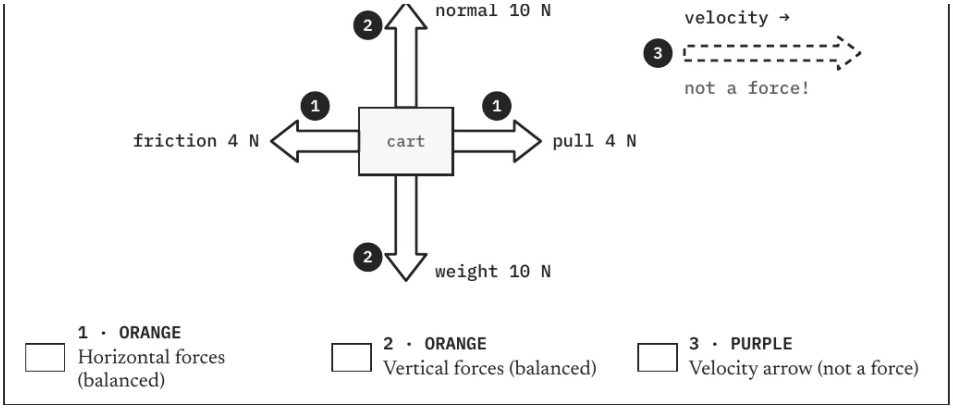
Also on your note page: Force, Net force, Newton's 1st law.

## B READ THE RELATIONSHIP

### Inertia means an object resists a change in velocity. More mass means more inertia.

Balanced forces → zero acceleration → the same velocity. Force: N. Mass: kg.

## C TRACE THE ORIGINAL FIGURE




Trace equal forces first. Then trace velocity. In pencil, label the arrows F or v.

## D FOLLOW THE WORKED EXAMPLE

### A cart rolls right, then slows down. What does that tell you about its net horizontal force?

**Given** moving right speed decreasing

**Rule** a ≠ 0 → net force ≠ 0

**Work** velocity changes → there is a net force; it opposes the motion

**Answer** Net force left opposite the motion

## E SAVE FOR THE FINAL MEETING

XJ is moving. A forward push is not needed to keep it moving. Gravity can still change its velocity.

# Read the physics: Inertia

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Push a cart across a rough floor, then let go. It slows because friction still acts on it. The floor and the cart interact. Inertia is a property of the cart; it does not push.

## 02 HOW IT WORKS

A force is a push or pull. Net force means all the forces on one object added with their directions. When the forces balance, acceleration is zero. A moving cart keeps its speed and direction. A resting cart stays at rest.

## 03 WHAT TO CHECK

Use a frame that is not speeding up or turning. A purple velocity arrow shows motion. Orange arrows show forces. Never add the purple arrow to the orange arrows.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

On the steady cart, 4 N right balances 4 N left. The table pushes up with 10 N, and Earth pulls down with 10 N. Now look at the second cart: it moves right but slows. Its velocity changes toward the left. Its net force points left.

## 05 CONNECT TO XJ

XJ needs no engine to keep moving. Over a short interval, use steady velocity as a model. A gravity pull can change the speed or direction. A new observation can test the model.

### SAY IT BEFORE YOU WRITE

Is the cart moving, or is its velocity changing? How can you tell? The cart moves ___. Its velocity ___ because the forces ___.

# Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

## A FRAYER MODEL

### Inertia / ES: inercia / PT: inércia

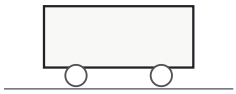
Say it in any language. Draw or use a few words. Use the notes to check your idea.

| Meaning | How can you tell? |
|---|---|
| To me, inertia means … | Same net force, same time. Both carts start at rest. Which gains less speed? The empty / loaded cart gains less because ___. |
| **Example or sketch** | **Non-example** |
| I see it when … | This looks close, but … |

## B DRAW WITH YOUR TEACHER

### Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.

Labels you can use: balanced · steady · changes.

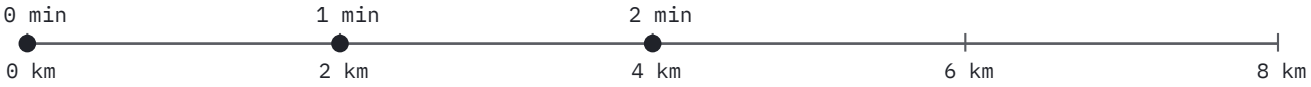


**Tell a partner:** The cart moves ___. Its velocity ___ because the forces ___.

# One camera. Which window?

SECTION E / BACK / FINAL MEETING ONLY Name _____

**XJ is drifting right. Tracker fixes: 0 min → 0 km; 1 min → +2 km; 2 min → +4 km. At minute 3, one camera can aim at only one window: A +4, B +6 or C +8 km.**




## PICK A JOB / DELIVER ONE PRODUCT

☐ **Tracking engineer**

Draw the predicted fix. Radio the operator: "At minute 3, aim at window ___." Use the spacing and the purple velocity arrow to explain your call.

☐ **Live science broadcaster**

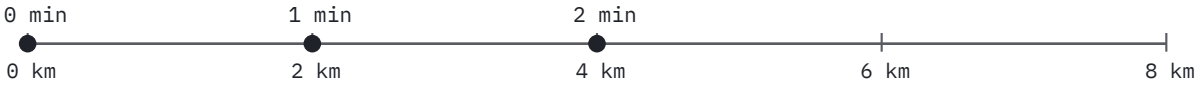
Deliver a 15-second update: tell viewers which window to watch. A caller asks, "No engine? Why won't it stop?" Use your cart drawing to answer on air.

**Words: aim • window • equal gaps • coast. Radio: "Aim at ___ because ___."**

On air: "Watch window ___. Without an engine, XJ can still ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE




## INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

**"We first said ___. Now we ___ because ___."**